

A. SPECIFIC AIMS

Infectious diseases caused by microbial pathogens in animals and humans contribute substantially to global mortality, morbidity, and economic loss, yet identifying the primary reservoir of infection remains a major challenge. For organisms such as *Escherichia coli*, which circulate between humans and food animals and can exist as both commensals and pathogens, current genomic approaches are poorly suited to resolving recent host transitions. Methods based on conserved core-genome variation (e.g., MLST and phylogenetics) reflect long-term evolutionary history but lack the resolution needed to infer the source of contemporary infections.

Our preliminary work demonstrates that accessory genome features, particularly mobile genetic elements (MGEs), can serve as informative indicators of host association. We demonstrated that markers of poultry-adapted *E. coli* such as ColV plasmids can resolve the likely reservoir of human clinical isolates within shared phylogenetic backgrounds. Building on this concept, we curated a pilot set of 17 host-associated MGEs associated with humans, chickens, turkey and pork and showed that these features can be incorporated into an epidemiological framework to perform source attribution of clinical *E. coli* isolates from a cohort in Arizona, USA. These findings supported a broader conceptual framework in which MGEs function as probabilistic “soft markers”: individually imperfect, but collectively capable of resolving host origin. My central hypothesis is that combinations of host-associated MGEs encode sufficient ecological signal to enable probabilistic inference of the host reservoir of *E. coli* isolates and estimate zoonotic burden in human populations. My overall objective is to develop a scalable, reproducible computational framework that identifies host-associated MGEs and integrates them into quantitative models for host attribution. My long-term goal is to enable high-resolution tracking of microbial transmission across the human–animal interface and support upstream interventions to reduce zoonotic disease burden.

Aim 1. Identify host-associated mobile genetic elements in *Escherichia coli*.

We will systematically identify new MGEs associated with human and food-animal reservoirs including beef-associated MGEs and additional MGEs for chicken, turkey, and pork using large-scale comparative genomics. We will (i) curate and harmonize a diverse genome collection; (ii) identify host-associated accessory genes using lineage-aware statistical models; (iii) assemble these genes into syntenic multigene networks representing candidate MGEs; and (iv) develop robust methods to detect these elements in new genomes despite structural variation. These efforts will generate a validated set of host-associated MGEs and a reproducible pipeline for their identification.

Expected outcome: A comprehensive catalog of host-associated MGEs in *E. coli* and a general scalable framework for detecting accessory genome signatures in microbes linked to specific host reservoirs.

Aim 2. Evaluate host-associated MGEs and an MGE-based probabilistic model for host attribution to estimate zoonotic burden in extraintestinal pathogenic *E. coli* (ExPEC) urinary tract infections.

To support robust estimation of zoonotic burden, we will test whether host-associated MGEs identified in our Arizona pilot study and in Aim 1, as well as the corresponding MGE-based host-attribution model, retain discriminatory value in independent cohorts. We will apply the framework to (a) in-house California collection of adult UTI and food-animal *E. coli* isolates (b) Washington, DC collection of pediatric UTI and food-animal *E. coli* isolates (c) publicly available *E. coli* genomes from National Antimicrobial Monitoring Systems (NARMS). We will evaluate (i) reproducibility of source-associated MGE prevalence, (ii) transportability of the probabilistic host-attribution model, and (iii) heterogeneity in generalization across cohorts, geography, age group, and source composition. We will then use the validated model to estimate the proportion of human ExPEC UTI isolates attributable to zoonotic reservoirs in pediatric and adult populations.

Expected outcome: This aim will identify which host-associated MGEs are broadly transferable across cohorts and determine whether MGE-based host attribution can provide robust, uncertainty-aware estimates of zoonotic burden in human UTI isolates. We will also as result estimate the extent to which zoonotic reservoirs contribute to pediatric and adult ExPEC infections.

Impact. Completion of these aims will provide broadly applicable tools for genomic epidemiology, enabling investigators to infer likely reservoirs, detect recent cross-species transmission events, and quantify zoonotic disease burden. This project will provide the trainee with advanced expertise in microbial genomics, statistical modeling, and reproducible bioinformatics, supporting the long-term goal of developing integrative deep learning driven methods for infectious disease surveillance.

SIGNIFICANCE

Zoonotic pathogens represent a major and under-resolved source of human infectious disease (1), yet identifying the primary reservoir of infection remains a fundamental challenge in genomic epidemiology. This gap is particularly evident for *E. coli*, a ubiquitous organism that circulates between humans and food animals (2–5) and can transition from commensal colonizer to invasive pathogen (6, 7). A substantial fraction of extraintestinal infections caused by *E. coli*, including urinary tract infections and bloodstream infections (8), may originate from animal reservoirs, but current methods lack the resolution to reliably detect and estimate zoonotic contribution.

Existing approaches to host attribution rely on conserved genomic features, including multilocus sequence typing (MLST) and core-genome phylogenetics (9, 10). While these methods are effective for defining population structure and reconstructing clonal relationships, they are poorly suited to identifying the precise host origin of *E. coli* strains from sporadic extraintestinal infections (11). Because *E. coli* lineages circulate across host species over extended time scales, core-genome similarity often reflects shared ancestry rather than recent sporadic transmission, obscuring the likely reservoir of infection.

This proposal addresses a critical gap by shifting the focus from conserved core-genome variation to dynamic accessory genome features, particularly MGEs, as indicators of host association. MGEs, including plasmids, transposons, and phage-derived regions, are frequently gained and lost during host adaptation and can retain signatures of prior ecological niches (12, 13). Preliminary work from our group demonstrates the feasibility of this approach: we have shown that ColV plasmids serve as markers of poultry-associated *E. coli* (14) and that incorporating accessory genome features into probabilistic models improves host attribution relative to lineage-based methods (15).

Despite the availability of large-scale genomic datasets, there are currently no scalable, reproducible frameworks to systematically identify host-associated MGEs or integrate them into quantitative models of host attribution and zoonotic burden estimation. This limitation prevents investigators from fully leveraging genomic data to understand cross-species transmission and to inform upstream interventions.

The proposed research directly addresses this need by identifying host-associated MGEs across human and food-animal reservoirs, developing computational tools to detect these signatures in new isolates, curating a database of host-associated markers, and testing both previously published and newly identified MGEs in independent cohorts to define a robust set of markers that generalize across geography and age groups. By enabling higher-resolution attribution of infections to likely reservoirs, this work will provide critical insight into the pathways through which pathogens move between animals and humans.

Impact: The ability to resolve the origin of infections at the human–animal interface has the potential to transform genomic surveillance and inform targeted interventions, including improved food production practices, source-directed control strategies, and prioritization of vaccine targets. More broadly, this framework will establish a generalizable approach for interpreting microbial genomes in an ecological context, with applications extending beyond *E. coli* to other zoonotic pathogens.

INNOVATION

This project is innovative because it redefines microbial source attribution by leveraging dynamic accessory genome features rather than conserved core-genome variation, enabling inference of recent host-associated ecological signal not captured by existing approaches.

First, this work introduces a conceptual shift in genomic epidemiology. Current attribution methods rely on lineage-based markers that reflect long-term evolutionary history but lack resolution for recent ecological exchange across hosts (16, 17). In contrast, this project establishes MGEs as probabilistic indicators of host association, exploiting their rapid turnover and ecological responsiveness to capture recent host adaptation events. This represents a fundamental departure from prevailing paradigms and enables a new class of inference grounded in accessory genome dynamics.

Second, this project develops a scalable and interpretable computational framework for transforming accessory genome variation into epidemiologically actionable features. Rather than treating genes as independent units, this approach reconstructs syntenic multigene networks that reflect biologically coherent mobile elements and integrates these features into probabilistic models of host attribution. This framework

bridges a critical gap between pangenome analysis and epidemiological inference, providing both predictive power and biological interpretability.

Third, this work will generate one of the first systematically curated, queryable resource linking mobile genetic elements to host reservoirs. By combining genomic, functional, and epidemiological data, this resource will enable automated detection of host-associated elements in new datasets and facilitate comparative analyses across host species and geographic regions. This transforms static genomic data into a reusable infrastructure for discovery and application.

Table 1. Existing food animal <i>E. coli</i> genomes					
Collection	Human clinical (UTI/BSI)	Chicken	Turkey	Pork	Beef
Total GWU Collection	3519	2161	1279	1147	780
Arizona	1118	1149	473	310	0
DC	52	42	14	0	0
California	2349	970	792	837	780
NARMS (world)	48375	25019	9262	17928	28035
NARMS (USA)	13185	7446	8038	6188	18217

Finally, this project advances beyond method development to enable new biological insight into how host-specific selective pressures shape microbial genomes. By contextualizing virulence factors, antimicrobial resistance genes, and other mobile elements within host environments, this work provides a foundation for understanding the mechanisms associated with host adaptation and zoonotic spread.

APPROACH

The central premise of this proposal is that accessory genome features, particularly MGEs, encode host-associated ecological signals that can support high-resolution source attribution. Our preliminary studies showed that poultry-adapted ColV plasmids(14) and a pilot set of 17 host-associated MGEs(15) can help resolve the likely reservoir of *E. coli* isolates that are closely related in the core genome. Building on this foundation, we will develop a scalable framework to systematically identify host-associated MGEs and integrate them into probabilistic models of host attribution. We will combine large-scale comparative genomics with lineage-aware statistical modeling to identify accessory genome features enriched in specific host populations, organize these features into biologically coherent multigene elements, and incorporate them into a Bayesian framework for host inference. This approach is designed to capture recent host transitions that are not captured by core-genome-based methods and to provide an interpretable, quantitative basis for attributing infections to their most likely source reservoirs.

Aim 1. Identify host-associated mobile genetic elements in *Escherichia coli*.

We will systematically identify host-associated MGEs linked to human and food-animal reservoirs, including newly resolved beef-associated MGEs and expanded marker sets for chicken, turkey, and pork, using large-scale comparative genomics. This aim will generate a validated catalog of host-associated MGEs and a reproducible framework for their identification and detection in new genomes.

1.1 Curate and harmonize a multi-host genome collection. We will assemble a comprehensive dataset of *E. coli* genomes from human and food-animal sources using existing collections ($n \approx 8,800$) after quality curation (≤ 100 contigs, $N50 \geq 50$ kb, Genome size ≥ 5 Mb) TABLE 1. Metadata, including host source, isolation context, sequence type (MLST)(18), and geographic origin, will be standardized across datasets. All genomes will be uniformly annotated to ensure consistency in downstream pangenome analyses.

1.2 Identify host-associated genes and assemble multigene MGE networks. Our previous approach contrasted human isolates with a broad food-animal category and yielded 17 host-associated MGEs. While that framework supported source attribution, it could not consistently distinguish among specific food-animal reservoirs. In this aim, we will expand the host-specific MGE panel to support discrimination among human, chicken, turkey, pig, and cattle-associated *E. coli*. We will identify accessory genes enriched in specific host populations Table 1 and organize them into biologically coherent mobile elements. MGE discovery will be restricted to the GWU collection, which provides a more controlled dataset for marker identification. Publicly available genomes from National Antimicrobial Monitoring Systems (NARMS) will be used only for external comparison and to assess the prevalence of identified MGEs outside the controlled discovery set.

Add a european collection on top of GWU (DANMAP or NethMAP)
preserve DANMAP for testing?

1.2.1. Pangenome construction and gene association. We will construct a pangenome with PPanGGOLiN(19), partition gene families into persistent and accessory components, and derive a synteny-based adjacency graph for downstream network reconstruction. Accessory gene families will be tested for host association using lineage-aware statistical models, including elastic net models implemented in pyseer (20) with cross-validation. Population structure will initially be controlled using MLST-based lineage definitions, with sensitivity analyses using alternative structure-aware encodings as needed. Genes reproducibly enriched in specific hosts will be retained as candidate host-associated features.

1.2.2. Network-based reconstruction of MGEs. Because MGEs function as multigene units rather than isolated genes, we will organize host-associated genes into syntenic gene networks to capture biologically meaningful structures of horizontal transfer. As detailed in figure 1, we will use the

pangenome-derived adjacency graph and will incorporate both direct adjacency between host-associated genes and indirect connections bridged by intermediate genes. To reduce redundancy and spurious grouping, gene-level relationships will be collapsed into cluster-level edges and filtered using host-prevalence criteria for intermediate clusters. Connected components in the resulting graph will define candidate multigene MGE networks, which will be annotated using PPanGGOLiN (19) co-evolving modules, gene composition, and structural properties such as size and edge density. This step converts individual gene associations into structured, interpretable genomic elements.

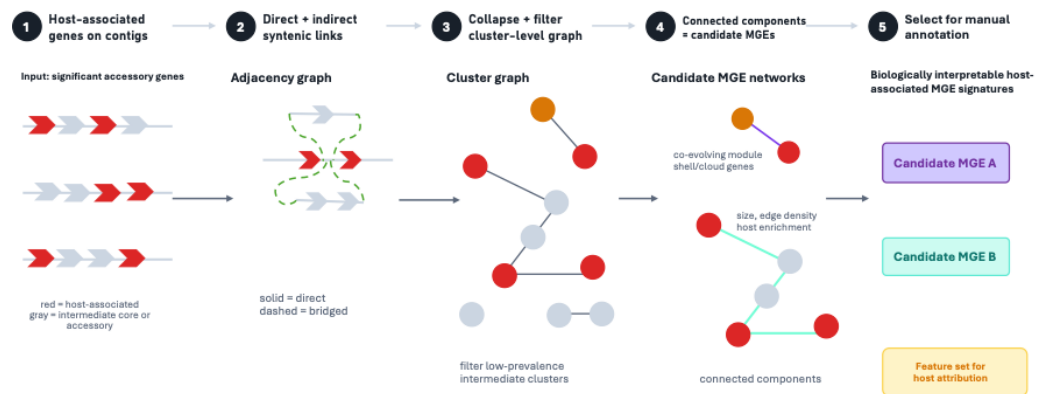


Fig 1. Network-based reconstruction of host-associated MGEs. Workflow for converting individual gene associations into biologically interpretable MGE networks that can be used as robust host-attribution features.

1.2.3. Quantification of host association For each candidate MGE network, we will quantify host association using complementary measures of prevalence and structural consistency across host populations. These will include overall prevalence, source-specific prevalence, synteny-based prevalence, and enrichment effect sizes relative to other host groups. We will also assess structural consistency within each network by quantifying the extent to which constituent genes remain co-localized across network-carrying genomes. Where appropriate, network-level enrichment will be formally tested across host populations. Host-associated MGEs will be prioritized as networks that show concordant host enrichment, stable syntenic organization, and reproducible detection across independent datasets. This multi-metric framework will help distinguish biologically meaningful MGEs from spurious or unstable gene associations.

1.2.4. Biological characterization of candidate MGEs.

Candidate networks will be evaluated for features characteristic of mobile elements (e.g., plasmids, prophages, transposases, integrative elements) using annotation tools (Text Box 1) and targeted curation.

This step will help prioritize biologically plausible MGEs for downstream detection and host-attribution modeling.

1.3 Develop methods to detect MGEs in new genomes.

For each MGE, we will develop detection models using features derived from gene composition and genomic context to identify complete or partial elements in new genomes. Depending on element structure and training signal, these models may take the form of statistical classifiers or rule-based detection algorithms. Confidence in MGE calls will be quantified using bootstrap-based

Text Box 1. Tools for manual identification of MGEs

- (i) Plasmids: Deepplasmid(21), mlplasmid(22), plasflow(23), PlasmidFinder (24), pMLST(25)
- (ii) Prophage: Phage Finder(26), PhiSpy(27)
- (iii) IS elements: ISfinder(28), **ISEScan(29)**
- (iv) Transposases: TnpPred(30)
- (v) Genomic Islands: islandPATH(31), AlienHunter(32), PAI-DB(33)
- (vi) Integrative and conjugative elements (ICEs) : ICEberg 2.0(34)
- (vii) Type IV Secretion systems: CONJscan (35), AtlasT4SS database(36)
- (viii) Conjugation genes: T346hunter(37)
- (ix) tRNA and tmRNA using tRNAscan (38), Aragorn (39)

procedures. Performance will be assessed using sensitivity, specificity, and robustness across independent datasets. **Talk to Jody about possible db (mySQL?)**

1.4 Create a queryable database of the host-associated MGEs We will develop a queryable database to support reproducibility, dissemination, and reuse of host-associated MGE discoveries. The database will store genomic, functional, and epidemiologic attributes of each MGE, including host source, prevalence, and annotations, and will support host-centric, species-centric, and geographically filtered queries. It will be implemented in a relational schema with API access to facilitate integration into external workflows.

1.5. Expected Outcomes and Interpretation. This aim will produce (i) a validated catalog of host-associated MGEs, (ii) a scalable and reproducible identification framework, and (iii) a robust feature set for probabilistic host attribution and (iv) a publicly accessible database for discovery and application. These outputs establish the foundation for modeling transmission across host populations in Aim 2.

1.6. Potential Pitfalls and Alternative Strategies. **1.6.1. Confounding by lineage or geography.** A key challenge is distinguishing true host association from confounding due to lineage or geography. Therefore, our analyses will incorporate lineage-aware models and be validated across independent datasets. Strong lineage-associated signals will be flagged for targeted evaluation. **1.6.2. Instability of individual gene markers.** To mitigate this, the framework emphasizes multigene networks and syntenic structure rather than single-gene associations. **1.6.3. Incomplete genome assemblies.** Genome assemblers are generally unable to assemble MGEs completely due to their plasticity. We do not expect genome assemblies to have assembled contigs that have all the genes in an MGE. Our developed detection methods will incorporate partial signatures and robustness analyses to ensure consistent performance across draft genomes. **1.6.4 Inability to train classifier.** If classifier-based MGE detection proves unstable or underpowered for some elements, we will implement rule-based calling schemes based on gene presence/absence patterns and element-level prevalence thresholds. This fallback strategy will preserve detection of complete and partial MGEs while avoiding overfitting.

Aim 2. Evaluate host-associated MGEs and an MGE-based probabilistic model for host attribution to estimate zoonotic burden in extraintestinal pathogenic *E. coli* (ExPEC) urinary tract infections.

Our preliminary studies showed that host-associated MGEs identified from *E. coli* collected from Arizona human clinical and retail meat samples can support probabilistic source attribution and distinguish extraintestinal *E. coli* associated with food-animal reservoirs from those associated with humans. However, the core evolutionary background of *E. coli* strains may vary across cohorts that differ in geography and source composition, potentially affecting portability of host-attribution models. The objective of this aim is to determine whether host-associated MGEs and an MGE-based Bayesian host-attribution framework generalize across independent cohorts and can be used to estimate zoonotic burden in human ExPEC urinary tract infections. Our working hypothesis is that although strain backgrounds will vary across vertebrate hosts and regions, a subset of host-associated MGEs will retain reproducible source-specific patterns, thereby supporting transportable probabilistic host attribution across studies. We will test this hypothesis in two stages. First, we will evaluate the previously published Arizona-derived panel of 17 host-associated MGEs (15) and its corresponding Bayesian attribution framework in independent California and Washington, DC cohorts. Second, we will evaluate an expanded MGE panel identified in Aim 1 using a broader labeled reference set and determine whether these additional markers improve host resolution and reduce uncertainty in zoonotic burden estimation. Together, these analyses will determine whether MGE-based host-attribution signals are transferable across cohorts and define how prediction uncertainty varies with geography, age structure, and source composition.

2.1 Cohorts and validation design **2.1.1 Study collections.** We will analyze *E. coli* isolates from three epidemiologically distinct study collections, together with publicly available genomes. The Arizona study, which served as the basis for the previously published 17-MGE framework, includes UTI isolates from children and adults collected from January 1 to December 31, 2012 from a local hospital in Flagstaff, Arizona, together with retail meat isolates obtained from local grocery stores. Independent external cohorts include an in-house California collection of adult UTI isolates and retail meat isolates collected from February 2017 to May 2021, and a Washington, DC collection of pediatric UTI isolates collected from residents aged 2 months to 17 years between October 2013 and March 2014, together with retail meat isolates from local grocery stores. Publicly available genomes from National Antimicrobial Monitoring Systems (NARMS) will be incorporated only in

analyses involving the expanded marker set. These cohorts differ in geography and food-source representation and therefore provide a stringent setting in which to evaluate generalizability of host-associated MGEs and the attribution framework.

2.1.2 Validation framework for the original Arizona-derived model. To evaluate portability of the previously published framework, we will use the Arizona cohort as the labeled reference set for the Bayesian attribution model. In this analysis, host class labels will be provided only for reference isolates from Arizona. California and Washington, DC isolates will be treated as unlabeled test isolates, and posterior host probabilities will be inferred from observed MGE and lineage features. This design provides a strict external evaluation of whether the original 17 host-associated MGEs and the corresponding Bayesian framework retain discriminatory value in cohorts that were not used in marker discovery or model development.

2.1.3 Validation framework for the expanded MGE panel. We will next evaluate an expanded host-associated MGE panel identified in Aim 1 using a broader labeled reference set composed of Arizona and California isolates with known host class labels. Washington, DC isolates and publicly available NARMS genomes will serve as independent unlabeled test cohorts. This design will allow us to determine whether the expanded marker set improves host resolution, extends source attribution to newly represented reservoirs such as beef, and reduces uncertainty in posterior host assignment relative to the original 17-MGE framework. The two test cohorts provide complementary forms of validation: the Washington, DC collection will assess portability in a clinically distinct pediatric population, whereas NARMS will provide an external public-data benchmark for broader source diversity and surveillance-context generalization.

2.1.4 Generalizability and transportability endpoints. Across both validation stages, we will evaluate generalizability at the marker level and transportability at the model level. Marker-level assessment will determine whether host-associated MGEs retain reproducible structural organization, prevalence, and source-enrichment patterns across cohorts. Model-level assessment will determine whether posterior host-attribution patterns remain well separated and interpretable in independent test isolates, and whether assignment confidence or uncertainty shifts across cohorts that differ in geography and source composition. These analyses will provide the foundation for subsequent estimation of the proportion of human ExPEC UTI isolates attributable to zoonotic reservoirs in pediatric and adult populations.

2.2 Assess structural and epidemiologic generalizability of host-associated MGEs. For each of the 17 Arizona-derived host-associated MGEs and the newly discovered MGEs from Aim 1, we will map constituent genes to genomes in the Arizona, California, Washington, DC, and, where applicable, NARMS collections using MMseqs2. Within each cohort, we will construct pairwise gene co-occurrence matrices to assess whether the internal structure of each MGE is preserved across study populations. We will quantify matrix-level concordance between Arizona and external cohorts using Mantel tests and will investigate cohort-specific deviations through gene-pair homogeneity testing with multiple-testing correction. In parallel, we will quantify source-specific prevalence and enrichment for each MGE within each cohort. MGEs that retain concordant structural organization together with reproducible source-enrichment patterns across cohorts will be prioritized as generalizable host-associated elements, whereas MGEs exhibiting substantial structural or epidemiologic heterogeneity will be flagged for sequence-level investigation and possible refinement.

2.3 Quantify cross-cohort stability of MGE-host association. We will apply the MGE detection classifier developed in Aim 1 to identify the presence or partial presence of Arizona-derived and newly discovered MGEs in each validation cohort. For each MGE-by-host combination, we will estimate cohort-specific prevalence and log odds ratios comparing a given host category with all remaining represented sources. We will then use a random-effects meta-analytic framework, treating each cohort as an independent stratum, to pool effect estimates and formally test cross-cohort heterogeneity. We will report pooled effect sizes, 95% confidence intervals, Cochran's Q statistic, and the I^2 statistic. An $I^2 < 25\%$ will be interpreted as low heterogeneity supporting stable host association; I^2 of 25-50% as moderate heterogeneity warranting cautious interpretation; and $I^2 > 75\%$ as substantial heterogeneity indicating that the MGE-host association is cohort-dependent. Source categories absent from a given cohort will be handled in a source-aware manner; for example, pork-associated MGEs will be evaluated only in cohorts where pork isolates are represented, and beef-associated MGEs will be evaluated primarily in California and in expanded reference analyses.

2.4 Assess transportability of the Arizona-derived Bayesian attribution model. The previously published Bayesian latent class model, using the Arizona cohort as the labeled reference set and the 17 Arizona-derived host-associated MGEs as features, will be applied to unlabeled isolates from the California and Washington, DC cohorts. For each isolate, the model will output posterior probabilities across the represented source categories. Transportability will be evaluated using discrimination together with measures of assignment confidence and uncertainty. Discrimination will be assessed using one-vs-rest area under the receiver operating characteristic curve (AUC) for each source category and by examining posterior probability separation between isolates from the expected source and alternative source groups after prediction. Assignment confidence and uncertainty will be quantified using the maximum posterior probability and entropy of the posterior source distribution for each isolate. These metrics will be compared with Arizona internal validation results to determine how uncertainty shifts when the model is applied to geographically and epidemiologically distinct cohorts. Because the California cohort includes beef isolates, which are not represented in the Arizona reference set, we will explicitly examine posterior distributions for beef isolates to determine whether the original model shows systematic misattribution or, alternatively, appropriate uncertainty in the presence of an unrepresented source.

2.5 Compare original and expanded host-attribution models. We will augment the original 17-MGE Bayesian latent class model with the newly discovered MGEs from Aim 1 and fit the expanded model using Arizona and California isolates as the labeled reference set. Washington, DC isolates and NARMS genomes will serve as independent unlabeled evaluation cohorts. For each isolate, the model will output posterior probabilities across the represented source categories. We will compare the original and expanded models using discrimination together with measures of assignment confidence and uncertainty. Discrimination will be assessed using one-vs-rest area under the receiver operating characteristic curve (AUC) for each source category and by examining posterior probability separation between isolates from the expected source and alternative source groups after prediction. Assignment confidence and uncertainty will be quantified using the maximum posterior probability and entropy of the posterior source distribution for each isolate. These comparisons will determine whether the expanded MGE panel improves source discrimination, increases confidence for represented host categories, and reduces uncertainty in external cohorts, with particular emphasis on attribution for beef and other food-animal reservoirs not adequately captured by the original Arizona-derived framework.

2.6 Estimate zoonotic burden in pediatric and adult ExPEC UTI cohorts. MGEs that demonstrate reproducible host association and improve attribution performance in Sections 2.2–2.5 will be incorporated into the final Bayesian latent class model. Using Arizona and California isolates as the labeled reference set, we will infer posterior host-species origin probabilities for human ExPEC UTI isolates in pediatric and adult cohorts, including the Washington, DC pediatric collection and the adult California collection. Zoonotic burden will be summarized as the proportion of human UTI isolates assigned high posterior probability of food-animal origin, and we will report burden estimates overall and stratified by cohort. We will quantify uncertainty in these estimates using posterior attribution distributions and will perform sensitivity analyses across alternative posterior probability thresholds for high-confidence attribution. NARMS will be used as an external evaluation dataset for attribution performance and robustness, but not as the primary basis for burden estimates in our in-house clinical cohorts.

Expected outcomes and interpretation

This aim will determine which host-associated MGEs are structurally and epidemiologically transferable across independent cohorts and whether an expanded MGE panel improves host attribution beyond the original Arizona-derived framework. It will establish how attribution performance and uncertainty vary across cohorts that differ in geography and source representation and will generate uncertainty-aware estimates of zoonotic burden in pediatric and adult ExPEC urinary tract infections. These results will clarify which MGE signals are broadly portable, which are cohort-dependent, and how source composition influences model behavior in real-world epidemiologic settings.

Potential pitfalls and alternative strategies

Several challenges may arise in this aim. First, some MGEs may exhibit cohort-specific prevalence or structural heterogeneity, limiting their portability across studies. In such cases, we will refine element

boundaries, re-examine sequence composition, and distinguish broadly transferable MGEs from population-dependent markers rather than forcing a single universal panel. Second, attribution performance may degrade when the original Arizona-derived model is applied to cohorts containing underrepresented or previously unseen sources such as beef. If so, we will (i) we will train a supplementary BLCM model incorporating beef as an additional source category using California isolates with known provenance and evaluate whether its inclusion improves human attribution specificity (ii) rely on the expanded marker set and explicitly interpret dispersed posterior assignments as evidence of appropriate model uncertainty rather than model failure. Finally, some source categories may be sparsely represented in individual cohorts, reducing precision of prevalence or burden estimates. We will address this through source-aware analyses, meta-analytic pooling where appropriate, and transparent reporting of uncertainty.

Timeline

		2023	2024	2025	2026			2027		
		Aug-Dec	Jan-Dec	Jan-Dec	Jan-May	Jun	July-Dec	Jan	Feb	March
Milestone						★ Proposal defense		★ Share Dissertation with committee	★ Dissertation defense	★ Final Submission
Aim 1 analysis										
Prepare manuscript	Host adaptation pipeline - Methods							★ Manuscript ready		
Aim 2 analysis										
Prepare manuscript	Pediatric UTI			★ Manuscript ready						
	California bill SB27				★ Manuscript ready					

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